## Status of this sheet

The substitution $t=\tan\frac{A}{2}$ (Weierstrass) is **standard RMO/INMO technique** and a bridge to integral calculus; every model below is derivable from Part I identities. Nothing here "can never appear in JEE" — historically JEE-style papers have used it in disguised form; the classification is only that it exceeds the syllabus sentence "Trigonometric functions, general solution, identities".

## The substitution and what it buys

Put $t=\tan\dfrac{A}{2}$. Then (for $A\neq\pi+2k\pi$):

$$\sin A=\frac{2t}{1+t^2},\qquad \cos A=\frac{1-t^2}{1+t^2},\qquad \tan A=\frac{2t}{1-t^2}.$$

SHORT DERIVATION

**Derivation — divide the double-angle forms by $\cos^2$.** $\sin A=2\sin\frac{A}{2}\cos\frac{A}{2}=\frac{2\tan\frac{A}{2}}{\sec^2\frac{A}{2}}=\frac{2t}{1+t^2}$ (using $\sec^2=1+\tan^2$). Similarly $\cos A=\cos^2\frac{A}{2}-\sin^2\frac{A}{2}=\frac{1-t^2}{1+t^2}$ after dividing by $\cos^2\frac{A}{2}$. Tangent: ratio of the two ✓. The point: **any rational expression in $\sin A,\cos A$ becomes a rational function of $t$** — trig equations become polynomial equations; trig integrals $\int R(\sin x,\cos x)\,dx$ become rational integrals (calculus toolkit).

WORKED EXAMPLE

**Worked. Solve $\sin x+\cos x=1$ for $0\le x<2\pi$ with** $\frac{2t}{1+t^2}+\frac{1-t^2}{1+t^2}=1\Rightarrow 2t+1-t^2=1+t^2\Rightarrow 2t^2-$ or $t=1$. $t=0\Rightarrow x=0$; $t=1\Rightarrow\frac{x}{2}=\frac{\pi}{4}\Rightarrow x=\frac{\pi}{2}$. The excluded point $x=\pi$ (where $\cos\frac{x}{2}=0$) gives $\sin\pi+\cos\pi=-1\neq 1$ ✓ nothing lost. Answer: $x\in\{0,\frac{\pi}{2}\}$ ✓.

COMMON TRAP

**The blind spot — $A=\pi$ (mod $2\pi$)**

$t=\tan\frac{A}{2}$ is undefined when $\frac{A}{2}=\frac{\pi}{2}+k\pi$, i.e. $A=\pi+2k\pi$. Every solution there is invisible to the substitution — **always test $A=\pi+2k\pi$ in the original equation** and add it if it works. For $A\in(-\pi,\pi)$ the map $A\mapsto t$ is a bijection onto $\mathbb{R}$ (this is the clean domain window for the substitution).

KEY RESULT

**When to use it.** (1) Equations rational in $\sin,\cos$ that resist factorisation; (2) proving identities by clearing denominators in $t$; (3) converting geometry (slope $=\tan\frac{\theta}{2}$) into algebra; (4) (forward reference) any integral $\int R(\sin x,\cos x)dx$. Counter-indication: when half-angle or product-to-sum is quicker — do the cheap tool first.

## Worked models

WORKED EXAMPLE

**Worked. Solve $\sin x+\cos x=1$ over all reals**

General solutions from the two families: $x=2n\pi$ (from $t=0$) and $x=\frac{\pi}{2}+2n\pi$ (from $t=1$), $n\in\mathbb{Z}$; the blind-spot test $x=\pi+2n\pi$ fails. Combined: $x=2n\pi$ or $x=\frac{\pi}{2}+2n\pi$ ✓ (leave as the two families).

WORKED EXAMPLE

**Worked. Express $\frac{1-\cos x}{\sin x}$ in $t$**

$\frac{1-\cos x}{\sin x}=\frac{1-\frac{1-t^2}{1+t^2}}{\frac{2t}{1+t^2}}=\frac{\frac{2t^2}{1+t^2}}{\frac{2t}{1+t^2}}=t=\tan\frac{x}{2}$ ✓ — consistent with the half-angle identity of Part I. Any rational $R(\sin x,\cos x)$ now reduces by $t$-clearing denominators; $1+t^2\neq 0$ is automatic for real $t$.

## Assumptions & conventions

A real with $A \neq \pi + 2k\pi$ under the substitution; the blind spot is checked separately in every problem; rational expressions in $\sin, \cos$ mean ratios of polynomials; all conversions verified back in the original equation.

## Status of this sheet

Products like $\prod \sin \frac{k\pi}{n}$ are **classic RMO/INMO algebra** obtained by combining Part I identities with complex numbers (chapter 5 of the main sequence supplies $z = re^{i\theta}$ and $|z|$; only elementary modulus facts are used).
Classification: beyond the syllabus sentence, but every step is derivable from supplied identities and the definition of $|1 - e^{i\theta}|$.

## The two flagship products

$$\prod_{k=1}^{n-1} \sin \frac{k\pi}{n} = \frac{n}{2^{n-1}}$$

$$\prod_{k=1}^{n} \cos \frac{k\pi}{2n+1} = \frac{1}{2^n}$$

**SHORT DERIVATION**

**Derivation — factor $z^n - 1$ at the roots of unity, s**

The $n$ roots of $z^n = 1$ are $e^{2\pi ik/n}$, $k = 0, \ldots, n-1$. So $z^n - 1 = (z-1)\prod_{k=1}^{n-1}(z - e^{2\pi ik/n})$. Cancel $z - 1$ ($\frac{z^n-1}{z-1} = 1 + z + \cdots + z^{n-1}$) and set $z = 1$: $n = \prod_{k=1}^{n-1}(1 - e^{2\pi ik/n})$. Now $|1 - e^{i\theta}| = 2\left|\sin \frac{\theta}{2}\right|$ (chord of a unit-circle arc; expand $|1 - \cos\theta - i\sin\theta|$). Taking moduli: $n = \prod 2\sin \frac{k\pi}{n} = 2^{n-1}\prod \sin \frac{k\pi}{n}$ (each $\sin \frac{k\pi}{n} > 0$ for $1 \le k \le n-1$) ✓. The cosine product comes from the same factorisation at $z = -1$.

**WORKED EXAMPLE**

**Worked. $\prod_{k=1}^{8} \sin \frac{k\pi}{7}$ and $\cos\frac{\pi}{7}\cos\frac{2\pi}{7}\cos\frac{3\pi}{7}$**

$n = 7$: product $= \frac{7}{2^6} = \frac{7}{64} = 0.109375$ ✓ (direct: $0.43388 \times 0.78183 \times 0.97493 \times 0.97493 \times$ ). Cosine product with $2n + 1 = 7$ ($n = 3$): $\frac{1}{2^3} = \frac{1}{8}$ ✓ (direct: $0.90097 \times 0.62349 \times 0.22252 = 0.125$).

## The pattern to internalise

**A product of trig values at rational multiples of $\pi$ is a roots-of-unity problem in disguise.** Second tool: when the angles come from a polynomial whose roots are $\tan$-values or the values themselves, **Vieta** hands you sums/products for free (see the $\tan 20^\circ \tan 40^\circ \tan 80^\circ = \sqrt{3}$ model of Part I sheet 8 — same idea with $\tan 3\theta$).

**WORKED EXAMPLE**

**Worked. Sine product for $n = 5$: $\sin 36^\circ \sin 72^\circ$?**

$\prod_{k=1}^{4} \sin \frac{k\pi}{5} = \frac{5}{16}$. Pair $\sin \frac{\pi}{5} = \sin 36^\circ$, $\sin \frac{2\pi}{5} = \sin 72^\circ$, $\sin \frac{3\pi}{5} = \sin 72^\circ$, $\sin \frac{4\pi}{5} = \sin 36^\circ$: product $= (\sin 36^\circ \sin 72^\circ)^2 = \frac{5}{16}$, so $\sin 36^\circ \sin 72^\circ = \frac{\sqrt{5}}{4} \approx 0.5590$ ✓ (direct: $0.58779 \times 0.95106 = 0.55902$).

**COMMON TRAP**

**Traps on this sheet**

- $\prod_{k=1}^{n-1}$ vs $\prod_{k=1}^{n}$: the sine product has $n-1$ factors (the $k=0$ factor would be $\sin 0 = 0$); count factors before quoting.
- $|1-e^{i\theta}| = 2|\sin\frac{\theta}{2}|$ — drop the absolute values only after checking positivity (done via $0 < \frac{k\pi}{n} < \pi$).
- The cosine product $\prod_{k=1}^{n}\cos\frac{k\pi}{2n+1} = 2^{-n}$ has $n$ factors and denominator $2n+1$ — a different index bookkeeping from the sine product.
- Claims here need complex arithmetic: state $e^{i\theta} = \cos\theta + i\sin\theta$ and $|z|$ facts explicitly in a solution write-up (assumptions below).

## Assumptions & conventions

Complex numbers used: $e^{i\theta} = \cos\theta + i\sin\theta$, $|zw| = |z||w|$, $|1-e^{i\theta}| = 2|\sin\frac{\theta}{2}|$ (proved above in modulus form). All products finite; $n \geq 2$ integer; positivity of $\sin\frac{k\pi}{n}$ for $1 \leq k \leq n-1$ used to drop absolute values.

## Status of this sheet

The telescoping series $\sum\sin k\theta$ and the sums $\sum\cot^2$, $\sum\csc^2$ are ***standard olympiad algebra***; all derive from Part I transformation formulae and the roots-of-Vieta technique of sheet 2. Models are derivable from supplied information; classified as extensions because they sit beyond the current syllabus sentence.

### Sums with arguments in arithmetic progression (telescoping)

**The trick:** multiply by $2\sin\frac{\theta}{2}$ and watch the product-to-sum formula telescope.

$$\sum_{k=1}^{n}\sin k\theta = \frac{\sin\frac{n\theta}{2}\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$$

$$\sum_{k=1}^{n}\cos k\theta = \frac{\sin\frac{n\theta}{2}\cos\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$$

**SHORT DERIVATION**

**Derivation — the $2\sin\frac{\theta}{2}$ multiplier**

$2\sin\frac{\theta}{2}\sin k\theta = \cos\left(k-\frac{1}{2}\right)\theta - \cos\left(k+\frac{1}{2}\right)\theta$. Summing over $k = 1\ldots n$ telescopes to $\cos\frac{\theta}{2} - \cos\left(n+\frac{1}{2}\right)\theta$. Then apply $\cos C - \cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$ with $C = \frac{\theta}{2}$, $D = (n+\frac{1}{2})\theta$; $= 2\sin\frac{(n+1)\theta}{2}\sin\frac{n\theta}{2}$. Divide by $2\sin\frac{\theta}{2}$ ✓. (Valid for $\sin\frac{\theta}{2} \neq 0$, i.e. $\theta \notin 2\pi\mathbb{Z}$; the excluded cases are computed directly as sums of 0/1.)

**WORKED EXAMPLE**

**Worked check.** $n = 3$, $\theta = \frac{\pi}{6}$

Direct: $\sin 30^\circ + \sin 60^\circ + \sin 90^\circ = 0.5 + 0.86603$. Formula: $\frac{\sin 45^\circ \sin 60^\circ}{\sin 15^\circ} = \frac{0.70711 \times 0.86603}{0.25882} = 2.36603$ ✓.

### Sums of squares at rational multiples of $\pi$ (roots + Vieta)

The numbers $t_k = \tan\frac{k\pi}{n}$, $k = 1, \ldots, n-1$, are the non-zero roots of a polynomial obtained from $\tan(n\theta)$ in terms of $t = \tan\theta$ (write $\tan n\theta = \frac{\Im(1+it)^n}{\Re(1+it)^n}$ and set the numerator to 0). Vieta then gives sums of powers.

$$\sum_{k=1}^{n-1}\cot^2\frac{k\pi}{n} = \frac{(n-1)(n-2)}{3}, \qquad \sum_{k=1}^{n-1}\csc^2\frac{k\pi}{n} = \frac{n^2-1}{3}.$$

SHORT DERIVATION

**Derivation sketch — cotangents as roots of the co**

$\cot^2\frac{k\pi}{n} = \frac{1}{t_k^2}$, and the symmetric identity $\sum\frac{1}{t_k^2} = \left(\sum\frac{1}{t_k}\right)^2 - 2\sum_{j<k}\frac{1}{t_j t_k}$ turns Vieta's data for the $t_k$ into the closed form $\frac{(n-1)(n-2)}{3}$. Then $\csc^2 = 1 + \cot^2$ adds $(n-1)$ terms of 1: $\sum\csc^2 = (n-1) + \frac{(n-1)(n-2)}{3} = \frac{(n-1)(n+1)}{3} =$ ✓.

WORKED EXAMPLE

**Worked checks.** $n = 3, 4, 5$

$\cot^2$: $n=3$: $\cot^2 60^\circ + \cot^2 120^\circ = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} = \frac{2\cdot 1}{3}$ ✓. $n=4$: $1 + 0 + 1 = 2 = \frac{3\cdot 2}{3}$ ✓. $n=5$: $4 = \frac{4\cdot 3}{3}$ ✓. $\csc^2$: $n=3$: $\frac{4}{3} + \frac{4}{3} = \frac{8}{3} = \frac{9-1}{3}$ ✓; $n=4$: $2 + 1 + 2 = 5 = \frac{15}{3}$ ✓; $n=5$: $8 = \frac{24}{3}$ ✓.

KEY RESULT

**Whenever a sum runs over trig values at equally spaced angles:** (1) look for a telescoping multiplier ($2\sin\frac{\theta}{2}$ for AP-argument sums); (2) look for the polynomial whose roots they are (Vieta for square/power sums); (3) pair symmetric terms ($k \leftrightarrow n-k$) to halve the work; (4) validate a small case numerically before the proof.

COMMON TRAP

**Traps on this sheet**

- The telescoping formulas divide by $\sin\frac{\theta}{2}$: state $\theta \notin 2\pi\mathbb{Z}$ (or handle those cases directly).
- $\sum\sin k\theta$ closed form: the numerator is $\sin\frac{n\theta}{2}\cdot\sin\frac{(n+1)\theta}{2}$ — order $(n+1)$ vs $n$ swapped between the two factors vs the cosine version (which has $\cos\frac{(n+1)\theta}{2}$).
- $\cot^2\frac{k\pi}{n}$ has a term at $k = \frac{n}{2}$ (when $n$ even) equal to 0 — include it (it is $\cot^2\frac{\pi}{2} = 0$).
- $\tan n\theta$'s polynomial in $t = \tan\theta$ has **degree** $n$ in the tangent-angle formula bookkeeping; only the roots $t_k$ with $\tan$ defined count.

## Assumptions & conventions

$n \geq 2$ integer; $\theta$ real with $\sin\frac{\theta}{2} \neq 0$ in series formulas; Vieta applied to real polynomials; numeric checks above computed in radians (converted to degrees for readability).

## Status of this sheet

**Important classification note.** The current JEE Advanced syllabus sentence for trigonometry is "Trigonometric functions. General solution of trigonometric equations. Identities." — it does **not** list properties of triangles or heights & distances. The formulae below are the **standard triangle toolkit** used across JEE Main-level papers, olympiads (RMO/INMO geometry-algebra), and later chapters (conics, 3D, vectors). They are added theory (not derivable from the identities alone — they encode triangle geometry), but every model built here only uses Part I identities plus the sine/cosine rules as stated. Do not assume they "can never appear"; assume they are simply outside the quoted syllabus sentence and learn them as toolkit.

## The four triangle formulae

For a triangle with sides $a, b, c$ opposite angles $A, B, C$, circumradius $R$, inradius $r$, semiperimeter $s = \frac{a+b+c}{2}$, area $\Delta$:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R, \qquad a^2 = b^2 + c^2 - 2bc\cos A$$

$$\Delta = \frac{1}{2}bc\sin A = \sqrt{s(s-a)(s-b)(s-c)}, \qquad r = \frac{\Delta}{s}$$

$$\tan\frac{A}{2} = \frac{r}{s-a}$$

**SHORT DERIVATION**

**Cosine rule and $\Delta = rs$ — one line each**

Cosine rule: drop a height $h = c\sin B$ from $C$; project: $a = b\cos C + c\cos B$-style coordinate computation gives $a^2 = b^2 + c^2 - 2bc\cos A$ (or expand $|\vec{B} - \vec{C}|^2$ in the vectors chapter). $\Delta = rs$: split the triangle into three $(r, s)$-shaped pieces — $\Delta = \frac{1}{2}ra + \frac{1}{2}rb + \frac{1}{2}rc = rs$ ✓. $\tan\frac{A}{2} = \frac{r}{s-a}$: the incircle tangent lengths are $s-a, s-b, s-c$ and half-angle right triangles give the ratio.

### Derived special values (pure identities, pentagon angles)

$$\sin 18^\circ = \frac{\sqrt{5}-1}{4} \approx 0.30902$$

$$\cos 36^\circ = \frac{\sqrt{5}+1}{4} \approx 0.80902$$

**SHORT DERIVATION**

**Derivation — double-angle, then quadratic**

Let $x = 18^\circ$, so $5x = 90^\circ$ and $2x = 90^\circ - 3x$, hence $\sin 2x = \cos 3x$. Expand: $2\sin x\cos x = 4\cos^3 x - 3\cos x$; with $\cos x \neq 0$: $2\sin x = 4\cos^2 x - 3 = 4(1 - \sin^2 x) - 3 = 1$. So $4\sin^2 x + 2\sin x - 1 = 0 \Rightarrow \sin x = \frac{-2+\sqrt{4+16}}{8}$ (positive root; $\sin 18^\circ > 0$) ✓. Then $\cos 36^\circ = 1 - 2\sin^2 18^\circ = \frac{\sqrt{5}+1}{4}$ ✓ (check numerically: 0.809017).

**WORKED EXAMPLE**

**Worked. Triangle with $A = 60^\circ$, $b = 5$, $c = 8$; find $a$**

Cosine rule: $a^2 = 25 + 64 - 2\cdot 5\cdot 8\cdot\frac{1}{2} = 49 \Rightarrow a = 7$.

$\Delta = \frac{1}{2}\cdot 5\cdot 8\cdot\frac{\sqrt{3}}{2} = 10\sqrt{3} \approx 17.32$.

Cross-check with Heron: $s = \frac{7+5+8}{2} = 10$, $\sqrt{10\cdot 3\cdot 5\cdot 2} = \sqrt{300} = 10\sqrt{3}$ ✓.

**WORKED EXAMPLE**

**Worked. $R$ and $r$ for the same triangle**

$2R = \frac{a}{\sin A} = \frac{7}{\sqrt{3}/2} = \frac{14}{\sqrt{3}} \Rightarrow R = \frac{7}{\sqrt{3}} \approx 4.041$.

$r = \frac{\Delta}{s} = \frac{10\sqrt{3}}{10} = \sqrt{3} \approx 1.732$. Check $\tan\frac{A}{2} = \tan 30^\circ = \frac{1}{\sqrt{3}}$ vs $\frac{r}{s-a} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$ ✓.

COMMON TRAP

**Traps — triangle toolkit**

- The sine rule gives an **ambiguous case** (SSA): $a = b\frac{\sin A}{\sin B}$ with $\sin B = \frac{b\sin A}{a}$ may admit two triangles ($B$ and $180^\circ - B$).
- $\Delta = \frac{1}{2}(\text{base})(\text{height})$ needs the height; $\frac{1}{2}bc\sin A$ needs the angle **between** the two sides.
- Heron's formula: $s$ is semiperimeter; writing $s = a + b + c$ is the classic slip.
- $\tan\frac{A}{2} = \frac{r}{s-a}$ requires the usual triangle convention ($s > a$); for degenerate/failing configurations state the inequality.
- $18^\circ$-family values ($\sin 18^\circ, \cos 36^\circ, \sin 54^\circ = \cos 36^\circ$ etc.): derive from the quadratic each time if memory fails — and pick the sign from the quadrant ($18^\circ$ is QI).

## Assumptions & conventions

Triangle = non-degenerate (positive angles summing to $\pi$, positive sides satisfying triangle inequalities); $R$ circumradius, $r$ inradius, $s$ semiperimeter, $\Delta$ area; special values derived from double-angle identities as shown; toolkit status per the note at the top of this sheet.